```
┌──(kali㉿kali)-[~]
└─$ mkdir -p ~/CIP-B103-Lab9/{evidence,working,exported,reports,screenshots,scripts}

┌──(kali㉿kali)-[~]
└─$ cd ~/CIP-B103-Lab9

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ pwd
/home/kali/CIP-B103-Lab9

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ find . -maxdepth 1 -type d -print
.
./working
./screenshots
./evidence
./scripts
./reports
./exported
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ sudo apt install -y aircrack-ng wireshark tshark foremost binwalk xz-utils
[sudo] password for kali:
wireshark is already the newest version (4.6.6-1).
tshark is already the newest version (4.6.6-1).
foremost is already the newest version (1.5.7-11+b2).
binwalk is already the newest version (2.4.3+dfsg1-2).
The following packages were automatically installed and are no longer required:
  libraw23t64  libvidstab1.1  openjdk-21-jre  openjdk-21-jre-headless
Use 'sudo apt autoremove' to remove them.

Upgrading:
  aircrack-ng  liblzma-dev  liblzma5  xz-utils

Summary:
  Upgrading: 4, Installing: 0, Removing: 0, Not Upgrading: 1713
  Download size: 1,297 kB / 1,998 kB
  Space needed: 59.4 kB / 23.4 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 xz-utils amd64 5.8.3-1 [740 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 aircrack-ng amd64 1:1.7+git20230807.4
```

```
Get:2 http://http.kali.org/kali kali-rolling/main amd64 aircrack-ng amd64 1:1.7+git20230807.4
bf83f1a-3 [557 kB]
Fetched 1,297 kB in 3s (445 kB/s)
(Reading database… 432542 files and directories currently installed.)
Preparing to unpack …/liblzma-dev_5.8.3-1_amd64.deb…
Unpacking liblzma-dev:amd64 (5.8.3-1) over (5.8.2-2)…
Preparing to unpack …/liblzma5_5.8.3-1_amd64.deb…
Unpacking liblzma5:amd64 (5.8.3-1) over (5.8.2-2)…
Setting up liblzma5:amd64 (5.8.3-1)…
(Reading database… 432542 files and directories currently installed.)
Preparing to unpack …/xz-utils_5.8.3-1_amd64.deb…
Unpacking xz-utils (5.8.3-1) over (5.8.2-2)…
Preparing to unpack …/aircrack-ng_1%3a1.7+git20230807.4bf83f1a-3_amd64.deb…
Unpacking aircrack-ng (1:1.7+git20230807.4bf83f1a-3) over (1:1.7+git20230807.4bf83f1a-2+b1)…
Setting up aircrack-ng (1:1.7+git20230807.4bf83f1a-3)…
Setting up xz-utils (5.8.3-1)…
Setting up liblzma-dev:amd64 (5.8.3-1)…
Processing triggers for libc-bin (2.42-13)…
Processing triggers for man-db (2.13.1-1)…
Processing triggers for kali-menu (2026.1.5)…
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ wget -O evidence/file.xz 'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz'
--2026-08-11 18:18:00--  https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.109.133, 185.199.110.133, 185.199.111.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.109.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 12837660 (12M) [application/octet-stream]
Saving to: 'evidence/file.xz'

evidence/file.xz        100%[===================>]  12.24M  1.34MB/s    in 9.4s

2026-08-11 18:18:11 (1.31 MB/s) - 'evidence/file.xz' saved [12837660/12837660]
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ ls -lh evidence/
total 13M
-rw-rw-r-- 1 kali kali 13M Aug 11 18:18 file.xz

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6  evidence/file.xz
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ cp --preserve=timestamps evidence/file.xz working/file_working.xz

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ unxz -k working/file_working.xz

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ file working/file_working
working/file_working: pcap capture file, microsecond ts (little-endian) - version 2.4 (802.11, capture length 65535)

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ sha256sum working/file_working.xz working/file_working | tee reports/working_hashes.txt
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6  working/file_working.xz
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d  working/file_working
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ tree
.
├── evidence
│   └── file.xz
├── exported
├── reports
│   ├── file_xz_sha256.txt
│   └── working_hashes.txt
├── screenshots
├── scripts
└── working
    ├── file_working
    └── file_working.xz

7 directories, 5 files

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ CAP=working/file_working

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ echo "$CAP"
working/file_working

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ capinfos "$CAP" | tee reports/capinfos.txt
File name:           working/file_working
File type:           Wireshark/tcpdump/ ... - pcap
File encapsulation:  IEEE 802.11 Wireless LAN
File timestamp precision:  microseconds (6)
Packet size limit:   file hdr: 65535 bytes
Number of packets:   45 k
File size:           14 MB
Data size:           13 MB
Capture duration:    272.969795 seconds
Earliest packet time: 2015-03-06 02:51:54.364525
Latest packet time:   2015-03-06 02:56:27.334320
```

```
Data byte rate:      49 kBps
Data bit rate:       395 kbps
Average packet size: 299.13 bytes
Average packet rate: 165 packets/s
SHA256:              c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d
SHA1:                cb929fd6e407c13639e3168b0f35f921788478d3
Strict time order:   False
Number of interfaces in file: 1
Interface #0 info:
                     Encapsulation = IEEE 802.11 Wireless LAN (20 - ieee-802-11)
                     Capture length = 65535
                     Time precision = microseconds (6)
                     Time ticks per second = 1000000
                     Number of stat entries = 0
                     Number of packets = 45169

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ tshark -r "$CAP" -q -z io,phs | tee reports/protocol_hierarchy.txt

===================================================================
Protocol Hierarchy Statistics
Filter:

frame                                    frames:45169 bytes:13511274
  wlan                                   frames:45169 bytes:13511274
    data                                 frames:15713 bytes:13104128
===================================================================

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ tshark -r "$CAP" -Y 'wlan' -T fields \
-e frame.number \
-e frame.time \
-e wlan.fc.type \
-e wlan.fc.subtype \
-e wlan.sa \
-e wlan.da \
-e wlan.bssid \
| head -n 100 | tee reports/wlan_frame_sample.tsv
1       2015-03-06T02:51:54.364526000-0500      1       11
2       2015-03-06T02:51:54.364527000-0500      1       11
3       2015-03-06T02:51:54.364525000-0500      1       11
4       2015-03-06T02:51:54.364526000-0500      1       11
```

5	2015-03-06T02:51:54.365550000-0500	1	11
6	2015-03-06T02:51:54.366063000-0500	1	11
7	2015-03-06T02:51:54.366062000-0500	1	11
8	2015-03-06T02:51:54.366061000-0500	1	11
9	2015-03-06T02:51:54.366063000-0500	1	11
10	2015-03-06T02:51:54.366061000-0500	1	11
11	2015-03-06T02:51:54.366064000-0500	1	11
12	2015-03-06T02:51:54.366063000-0500	1	11
13	2015-03-06T02:51:54.366063000-0500	1	11
14	2015-03-06T02:51:54.366063000-0500	1	11
15	2015-03-06T02:51:54.366064000-0500	1	11
16	2015-03-06T02:51:54.366062000-0500	1	11
17	2015-03-06T02:51:54.366575000-0500	1	11
18	2015-03-06T02:51:54.366575000-0500	1	11
19	2015-03-06T02:51:54.366573000-0500	1	11
20	2015-03-06T02:51:54.366574000-0500	1	11
21	2015-03-06T02:51:54.366575000-0500	1	11
22	2015-03-06T02:51:54.366575000-0500	1	11
23	2015-03-06T02:51:54.366575000-0500	1	11
24	2015-03-06T02:51:54.366574000-0500	1	11
25	2015-03-06T02:51:54.366574000-0500	1	11

26	2015-03-06T02:51:54.366574000-0500	1	11
27	2015-03-06T02:51:54.366574000-0500	1	11
28	2015-03-06T02:51:54.366574000-0500	1	11
29	2015-03-06T02:51:54.366574000-0500	1	11
30	2015-03-06T02:51:54.366575000-0500	1	11
31	2015-03-06T02:51:54.366573000-0500	1	11
32	2015-03-06T02:51:54.366575000-0500	1	11
33	2015-03-06T02:51:54.366574000-0500	1	11
34	2015-03-06T02:51:54.366575000-0500	1	11
35	2015-03-06T02:51:54.366575000-0500	1	11
36	2015-03-06T02:51:54.366574000-0500	1	11
37	2015-03-06T02:51:54.366575000-0500	1	11
38	2015-03-06T02:51:54.366576000-0500	1	11
39	2015-03-06T02:51:54.366575000-0500	1	11
40	2015-03-06T02:51:54.366574000-0500	1	11
41	2015-03-06T02:51:54.366573000-0500	1	11
42	2015-03-06T02:51:54.366573000-0500	1	11
43	2015-03-06T02:51:54.366574000-0500	1	11
44	2015-03-06T02:51:54.367085000-0500	1	11
45	2015-03-06T02:51:54.367086000-0500	1	11
46	2015-03-06T02:51:54.367087000-0500	1	11

```
47      2015-03-06T02:51:54.367089000-0500      1       13
48      2015-03-06T02:51:54.367085000-0500      1       11
49      2015-03-06T02:51:54.367087000-0500      1       11
50      2015-03-06T02:51:54.367087000-0500      1       11
51      2015-03-06T02:51:54.367087000-0500      1       11
52      2015-03-06T02:51:54.367087000-0500      1       11
53      2015-03-06T02:51:54.367086000-0500      1       11
54      2015-03-06T02:51:54.367633000-0500      0       8       00:26:66:55:97:d6       ff:ff
:ff:ff:ff:ff    00:26:66:55:97:d6
55      2015-03-06T02:51:54.368111000-0500      1       11
56      2015-03-06T02:51:54.368111000-0500      1       11
57      2015-03-06T02:51:54.368110000-0500      1       11
58      2015-03-06T02:51:54.368110000-0500      1       11
59      2015-03-06T02:51:54.368110000-0500      1       11
60      2015-03-06T02:51:54.368110000-0500      1       11
61      2015-03-06T02:51:54.368622000-0500      1       11
62      2015-03-06T02:51:54.368623000-0500      1       11
63      2015-03-06T02:51:54.368621000-0500      1       11
64      2015-03-06T02:51:54.368622000-0500      1       11
65      2015-03-06T02:51:54.368621000-0500      1       11
66      2015-03-06T02:51:54.368623000-0500      1       11
```

```
67      2015-03-06T02:51:54.368622000-0500      1       11
68      2015-03-06T02:51:54.368623000-0500      1       11
69      2015-03-06T02:51:54.368622000-0500      1       11
70      2015-03-06T02:51:54.368622000-0500      1       11
71      2015-03-06T02:51:54.368623000-0500      1       11
72      2015-03-06T02:51:54.368621000-0500      1       11
73      2015-03-06T02:51:54.368622000-0500      1       11
74      2015-03-06T02:51:54.369134000-0500      1       11
75      2015-03-06T02:51:54.369134000-0500      1       11
76      2015-03-06T02:51:54.369135000-0500      1       11
77      2015-03-06T02:51:54.369648000-0500      1       11
78      2015-03-06T02:51:54.369646000-0500      1       11
79      2015-03-06T02:51:54.370159000-0500      1       11
80      2015-03-06T02:51:54.371182000-0500      1       11
81      2015-03-06T02:51:54.371181000-0500      1       11
82      2015-03-06T02:51:54.371183000-0500      1       11
83      2015-03-06T02:51:54.371694000-0500      1       11
84      2015-03-06T02:51:54.371694000-0500      1       11
85      2015-03-06T02:51:54.372207000-0500      1       11
86      2015-03-06T02:51:54.372718000-0500      1       11
87      2015-03-06T02:51:54.375279000-0500      1       11
```

```
88      2015-03-06T02:51:54.375791000-0500      1       11
89      2015-03-06T02:51:54.376813000-0500      1       11
90      2015-03-06T02:51:54.376815000-0500      1       11
91      2015-03-06T02:51:54.376815000-0500      1       11
92      2015-03-06T02:51:54.376815000-0500      1       11
93      2015-03-06T02:51:54.376814000-0500      1       11
94      2015-03-06T02:51:54.376815000-0500      1       11
95      2015-03-06T02:51:54.376814000-0500      1       11
96      2015-03-06T02:51:54.381421000-0500      1       11
97      2015-03-06T02:51:54.381423000-0500      1       11
98      2015-03-06T02:51:54.382446000-0500      1       11
99      2015-03-06T02:51:54.382446000-0500      1       11
100     2015-03-06T02:51:54.382957000-0500      1       11

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ tree
.
├── evidence
│   └── file.xz
├── exported
└── reports
```

```
└─$ tree
.
├── evidence
│   └── file.xz
├── exported
├── reports
│   ├── capinfos.txt
│   ├── file_xz_sha256.txt
│   ├── protocol_hierarchy.txt
│   ├── wlan_frame_sample.tsv
│   └── working_hashes.txt
├── screenshots
├── scripts
└── working
    ├── file_working
    └── file_working.xz

7 directories, 8 files

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields \
-e frame.number \
-e frame.time_epoch \
-e wlan.sa \
-e wlan.da \
-e wlan.bssid \
-e wlan.wep.iv \
-e wlan.wep.key \
| head -n 200 | tee reports/wep_protected_frames.tsv
9432    1425628319.232982000    00:26:66:55:97:d4       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×67ffec        0
10130   1425628320.138252000    f0:f6:1c:68:96:7c       00:26:66:55:97:d6       00:26:66:55:9
7:d6    0×c2cbeb        0
10155   1425628320.176656000    f0:f6:1c:68:96:7c       04:8d:38:48:e5:b4       00:26:66:55:9
7:d6    0×c3cbeb        0
10157   1425628320.177168000    f0:f6:1c:68:96:7c       04:8d:38:48:a8:b5       00:26:66:55:9
7:d6    0×c4cbeb        0
10159   1425628320.177168000    f0:f6:1c:68:96:7c       08:10:77:92:7c:2f       00:26:66:55:9
7:d6    0×c5cbeb        0
10161   1425628320.177168000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×c6cbeb        0
10163   1425628320.177168000    f0:f6:1c:68:96:7c       c8:3a:35:58:c4:ba       00:26:66:55:9
7:d6    0×c7cbeb        0
10165   1425628320.177167000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×c8cbeb        0
10167   1425628320.177686000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×68ffec        0
10170   1425628320.180760000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×69ffec        0
10171   1425628320.182288000    f0:f6:1c:68:96:7c       33:33:00:00:00:02       00:26:66:55:9
7:d6    0×c9cbeb        0
10172   1425628320.187417000    00:26:66:55:97:d4       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×6affec        0
10173   1425628320.187408000    f0:f6:1c:68:96:7c       33:33:00:00:00:02       00:26:66:55:9
7:d6    0×c9cbeb        0
10175   1425628320.187919000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×cacbeb        0
10177   1425628320.188952000    f0:f6:1c:68:96:7c       33:33:00:00:00:02       00:26:66:55:9
7:d6    0×6bffec        0
10178   1425628320.189976000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×6cffec        0
10183   1425628320.192016000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×cbcbeb        0
10185   1425628320.193048000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×6dffec        0
10224   1425628320.259088000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×cccbeb        0
10228   1425628320.262680000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×6effec        0
10229   1425628320.262678000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×6fffec        0
10233   1425628320.263184000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×cdcbeb        0
10261   1425628320.419350000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×cecbeb        0
10266   1425628320.420374000    f0:f6:1c:68:96:7c       01:00:5e:00:00:02       00:26:66:55:9
7:d6    0×cfcbeb        0
10268   1425628320.420374000    f0:f6:1c:68:96:7c       01:00:5e:00:00:fb       00:26:66:55:9
7:d6    0×d0cbeb        0
10270   1425628320.420886000    f0:f6:1c:68:96:7c       01:00:5e:00:00:02       00:26:66:55:9
7:d6    0×70ffec        0
```

```
10272   1425628320.422421000    f0:f6:1c:68:96:7c       01:00:5e:00:00:fb       00:26:66:55:9
7:d6    0×71ffec        0
10353   1425628320.583190000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×72ffec        0
10355   1425628320.583700000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×73ffec        0
10357   1425628320.583702000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×d1cbeb        0
10359   1425628320.584726000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×74ffec        0
10360   1425628320.586774000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×d2cbeb        0
10367   1425628320.789014000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×d3cbeb        0
10375   1425628320.936471000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×d4cbeb        0
10377   1425628320.936469000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×75ffec        0
10379   1425628320.937495000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×76ffec        0
10380   1425628320.962071000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
```

```
7:d6    0×d5cbeb        0
10382   1425628320.976407000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×d6cbeb        0
10388   1425628321.112150000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×77ffec        0
10390   1425628321.139798000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×d7cbeb        0
10392   1425628321.156181000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×d8cbeb        0
10394   1425628321.157208000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×78ffec        0
10401   1425628321.287763000    f0:f6:1c:68:96:7c       33:33:00:00:00:16       00:26:66:55:9
7:d6    0×d9cbeb        0
10403   1425628321.287766000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×79ffec        0
10405   1425628321.287766000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×7affec        0
10407   1425628321.287767000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×7bffec        0
10409   1425628321.288279000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×7cffec        0
```

```
10411   1425628321.289816000    f0:f6:1c:68:96:7c       33:33:00:00:00:16       00:26:66:55:9
7:d6    0×7dffec        0
10412   1425628321.321554000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×dacbeb        0
10414   1425628321.325143000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×7effec        0
10415   1425628321.327698000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×dbcbeb        0
10417   1425628321.327698000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×dccbeb        0
10419   1425628321.328214000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×7fffec        0
10426   1425628321.462355000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×ddcbeb        0
10431   1425628321.488979000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×decbeb        0
10433   1425628321.490008000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×80ffec        0
10446   1425628321.813588000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×81ffec        0
10448   1425628321.841235000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
```

```
7:d6    0×dfcbeb        0
10458   1425628322.163860000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×82ffec        0
10460   1425628322.163860000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×83ffec        0
10462   1425628322.193046000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×e0cbeb        0
10464   1425628322.194070000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×e1cbeb        0
10476   1425628322.516627000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×84ffec        0
10478   1425628322.544276000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×e2cbeb        0
10480   1425628322.545302000    f0:f6:1c:68:96:7c       ff:ff:ff:ff:ff:ff       00:26:66:55:9
7:d6    0×85ffec        0
10481   1425628322.545300000    00:26:66:55:97:d4       f0:f6:1c:68:96:7c       00:26:66:55:9
7:d6    0×86ffec        0
10483   1425628322.546325000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×e3cbeb        0
10485   1425628322.547860000    f0:f6:1c:68:96:7c       00:26:66:55:97:d4       00:26:66:55:9
7:d6    0×e4cbeb        0
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields \
-e wlan.wep.iv \
| sort | uniq -c | sort -nr | head -n 30 \
| tee reports/repeated_iv_summary.txt
      7 0×42e8eb
      7 0×28d3eb
      6 0×95dfeb
      5 0×9415ed
      5 0×86e5eb
      5 0×7ae5eb
      5 0×28e7eb
      5 0×19e5eb
      4 0×d5d7eb
      4 0×d3e8eb
      4 0×c915ed
      4 0×c7d8eb
      4 0×ade4eb
      4 0×73e5eb
      4 0×65d8eb
```

```
      4 0×ade4eb
      4 0×73e5eb
      4 0×65d8eb
      4 0×4ad5eb
      4 0×31e7eb
      4 0×2723ed
      4 0×10e7eb
      3 0×ffe5eb
      3 0×f609ed
      3 0×a307ed
      3 0×97dbeb
      3 0×96d1eb
      3 0×92e8eb
      3 0×91e8eb
      3 0×8adeeb
      3 0×630aed
      3 0×56e8eb
      3 0×3c1aed

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ aircrack-ng "$CAP" | tee reports/aircrack_output.txt
Reading packets, please wait...
Opening working/file_working
Read 45169 packets.            Got 15477 out of 15000 IVsStarting PTW attack with 15477 ivs.

   #  BSSID              ESSID                     Encryption

   1  00:26:66:55:97:D6  cgnetwork                 WEP (15477 IVs)

Choosing first network as target.

Reading packets, please wait...
Opening working/file_working
Read 45169 packets.

1 potential targets

Attack will be restarted every 5000 captured ivs.
```

```
                                 Aircrack-ng 1.7

                     [00:00:00] Tested 83 keys (got 15477 IVs)

   KB    depth   byte(vote)
    0    0/  1   A4(22784) 62(20992) A8(19968) B6(19968) 42(19456) 6E(19456)
    1    0/  1   3D(23040) 51(20736) 07(20480) 62(19968) 7B(19968) 1F(19712)
    2    0/  1   F6(23808) E4(20992) D0(20736) 68(20224) 95(19712) 38(19456)
    3    1/ 10   F3(20480) C5(19968) D0(19968) 3E(19712) 43(19456) 52(19456)
    4    6/  9   01(19712) 20(19456) 3E(19456) 52(19456) 5C(19456) 90(19456)

                        KEY FOUND! [ A4:3D:F6:F3:74 ]
        Decrypted correctly: 100%


┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ cat reports/aircrack_output.txt
Reading packets, please wait...
Opening working/file_working
```

```
Read 45169 packets.            Got 15477 out of 15000 IVsStarting PTW attack with 15477 ivs.

   #  BSSID              ESSID                     Encryption

   1  00:26:66:55:97:D6  cgnetwork                 WEP (15477 IVs)

Choosing first network as target.

Reading packets, please wait...
Opening working/file_working
Read 45169 packets.

1 potential targets

Attack will be restarted every 5000 captured ivs.

                                 Aircrack-ng 1.7


                     [00:00:00] Tested 83 keys (got 15477 IVs)
```

```
 KB    depth   byte(vote)
  0    0/  1   A4(22784) 62(20992) A8(19968) B6(19968) 42(19456) 6E(19456)
  1    0/  1   3D(23040) 51(20736) 07(20480) 62(19968) 7B(19968) 1F(19712)
  2    0/  1   F6(23808) E4(20992) D0(20736) 68(20224) 95(19712) 38(19456)
  3    1/ 10   F3(20480) C5(19968) D0(19968) 3E(19712) 43(19456) 52(19456)
  4    6/  9   01(19712) 20(19456) 3E(19456) 52(19456) 5C(19456) 90(19456)

                      KEY FOUND! [ A4:3D:F6:F3:74 ]
      Decrypted correctly: 100%


┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ cat reports/validated_wep40_key_masked.txt
cat: reports/validated_wep40_key_masked.txt: No such file or directory

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ printf 'A4:3D:F6:F3:74\n' | tee reports/validated_wep40_key_masked.txt
A4:3D:F6:F3:74

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ ls -l reports/validated_wep40_key_masked.txt
-rw-rw-r-- 1 kali kali 15 Aug 11 18:51 reports/validated_wep40_key_masked.txt

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ cat reports/validated_wep40_key_masked.txt
A4:3D:F6:F3:74

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ ls -l reports/aircrack_output.txt
-rw-rw-r-- 1 kali kali 1058 Aug 11 18:49 reports/aircrack_output.txt

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ grep -E 'KEY FOUND|Decrypted correctly|BSSID|ESSID|WEP' reports/aircrack_output.txt
   #  BSSID              ESSID                     Encryption
   1  00:26:66:55:97:D6  cgnetwork                 WEP (15477 IVs)
                         KEY FOUND! [ A4:3D:F6:F3:74 ]
        Decrypted correctly: 100%

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ cd ~/CIP-B103-Lab9/working

┌──(kali㉿kali)-[~/CIP-B103-Lab9/working]
└─$ pwd
/home/kali/CIP-B103-Lab9/working

┌──(kali㉿kali)-[~/CIP-B103-Lab9/working]
└─$ airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
Total number of stations seen            10
Total number of packets read          45169
Total number of WEP data packets      15477
Total number of WPA data packets          0
Number of plaintext data packets          0
Number of decrypted WEP  packets      15477
Number of corrupted WEP  packets          0
Number of decrypted WPA  packets          0
Number of bad TKIP (WPA) packets          0
Number of bad CCMP (WPA) packets          0
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9/working]
└─$ cd ..

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt
   526448  13904 -rw-rw-r--    1 kali     kali     14234002 Aug 11 18:18 working/file_working
   526458  12488 -rw-rw-r--    1 kali     kali     12784501 Aug 11 18:58 working/file_working-
dec
   526447  12540 -rw-rw-r--    1 kali     kali     12837660 Aug 11 18:18 working/file_working.
xz

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ sha256sum working/* | tee reports/all_working_file_hashes.txt
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d  working/file_working
167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04  working/file_working-dec
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6  working/file_working.xz

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ DEC=$(find working -maxdepth 1 -type f -name '*-dec*' | head -n 1)

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ echo "Decrypted capture: $DEC"
Decrypted capture: working/file_working-dec

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ echo "Decrypted capture: $DEC" | tee reports/decrypted_capture_path.txt
Decrypted capture: working/file_working-dec

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ tshark -r "$DEC" -q -z endpoints,eth | tee reports/ethernet_endpoints.txt
================================================================================
Ethernet Endpoints
Filter:<No Filter>
                      | Packets | |  Bytes  | | Tx Packets | | Tx Bytes | | Rx Packets | | R
x Bytes |
Apple_68:96:7c             15407   12 MB           7076      616 kB           8331      11
 MB
EFMNetworks_55:97:d4       15020   12 MB           8178      11 MB            6842      59
3 kB
ASUSTekCOMPU_2a:c2:7a        358   281 kB           223       270 kB           135       1
1 kB
IPv4mcast_fb                  42   6,648 bytes        0      0 bytes           42       6,
648 bytes
IPv6mcast_fb                  36   7,022 bytes        0      0 bytes           36       7,
022 bytes
Broadcast                     31   3,386 bytes        0      0 bytes           31       3,
386 bytes
IPv4mcast_7f:ff:fa            31   11 kB              0      0 bytes           31       11
 kB
IPv6mcast_16                  10   920 bytes          0      0 bytes           10       92
0 bytes
IPv6mcast_02                   9   630 bytes          0      0 bytes            9       63
0 bytes
IPv4mcast_16                   3   180 bytes          0      0 bytes            3       18
0 bytes
IPv4mcast_02                   2   92 bytes           0      0 bytes            2       92
 bytes
NetisTechnol_48:e5:b4          1   42 bytes           0      0 bytes            1       4
```

```
2 bytes
NetisTechnol_48:a8:b5            1   42 bytes            0      0 bytes              1      4
2 bytes
08:10:77:92:7c:2f                1   42 bytes            0      0 bytes              1      42
 bytes
TendaTechnol_58:c4:ba            1   42 bytes            0      0 bytes              1      4
2 bytes
IPv6mcast_ff:2a:c2:7a            1   78 bytes            0      0 bytes              1      7
8 bytes
═══════════════════════════════════════════════════════════════════════════════

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ tshark -r "$DEC" -q -z endpoints,ip | tee reports/ip_endpoints.txt
═══════════════════════════════════════════════════════════════════════════════
IPv4 Endpoints
Filter:<No Filter>
                       | Packets | |  Bytes  | | Tx Packets | | Tx Bytes | | Rx Packets | | R
x Bytes |
192.168.0.15               15309   12 MB            6995      608 kB            8314      11
 MB
198.90.20.111               8057   6,916 kB         4446      6,637 kB          3611      27
```

```
1 kB
192.168.0.9                  329   277 kB            197      265 kB             132      11
 kB
173.194.127.141              153   109 kB             80      102 kB              73      7,
364 bytes
173.194.127.55               143   89 kB              71      82 kB               72      6,
990 bytes
8.8.8.8                       81   10 kB              40      6,960 bytes         41      3,
391 bytes
199.16.158.186                66   15 kB              29      10 kB               37      5,
214 bytes
173.194.127.227               66   24 kB              26      15 kB               40      8,
455 bytes
173.194.127.229               55   22 kB              23      13 kB               32      9,
731 bytes
31.13.70.1                    51   19 kB              26      15 kB               25      3,
151 bytes
173.241.248.220               44   6,848 bytes        20      3,122 bytes         24      3,
726 bytes
198.41.208.140                44   28 kB              21      25 kB               23      2,
718 bytes
```

```
224.0.0.251                   42   6,648 bytes         0      0 bytes             42      6,
648 bytes
173.194.127.237               36   9,526 bytes        14      6,435 bytes         22      3,
091 bytes
216.58.220.234                34   15 kB              16      12 kB               18      2,
497 bytes
192.168.0.1                   33   12 kB              33      12 kB                0      0
bytes
173.241.248.219               33   5,733 bytes        14      2,807 bytes         19      2,
926 bytes
104.74.158.12                 33   21 kB              18      20 kB               15      1,
236 bytes
74.125.203.156                32   20 kB              16      17 kB               16      2,
534 bytes
239.255.255.250               31   11 kB               0      0 bytes             31      11
 kB
173.194.127.231               31   6,816 bytes        12      5,197 bytes         19      1,
619 bytes
198.41.209.137                28   16 kB              13      14 kB               15      1,
556 bytes
17.172.239.31                 25   6,504 bytes        12      3,973 bytes         13      2,
```

```
531 bytes
173.194.72.95                23   4,077 bytes          9      2,232 bytes          14       1,
845 bytes
173.194.127.122              21   12 kB               11      11 kB                10       1,
759 bytes
69.59.169.185                18   5,644 bytes          7      3,675 bytes          11       1,
969 bytes
72.246.103.16                16   3,697 bytes          6      1,732 bytes          10       1,
965 bytes
23.23.127.192                15   3,383 bytes          6      1,370 bytes           9       2,
013 bytes
173.194.127.198              14   4,480 bytes          6      2,931 bytes           8       1,
549 bytes
173.194.122.24               14   1,724 bytes          5      688 bytes             9       1,
036 bytes
23.21.133.99                 13   2,314 bytes          5      880 bytes             8       1,
434 bytes
50.31.251.34                 12   3,483 bytes          6      2,777 bytes           6       70
6 bytes
108.168.254.139              11   2,655 bytes          5      1,690 bytes           6       96
5 bytes
```

```
108.168.254.138              11   2,333 bytes          5      1,124 bytes           6       1,
209 bytes
54.240.248.207               10   1,217 bytes          4      441 bytes             6       77
6 bytes
173.194.127.100              10   1,921 bytes          4      715 bytes             6       1,
206 bytes
54.169.42.244                10   2,054 bytes          4      1,054 bytes           6       1,
000 bytes
173.194.127.119              10   1,534 bytes          4      670 bytes             6       86
4 bytes
173.194.127.128               9   1,468 bytes          3      604 bytes             6       86
4 bytes
23.21.196.154                 8   975 bytes            2      140 bytes             6       83
5 bytes
203.190.124.19                8   1,742 bytes          3      648 bytes             5       1,
094 bytes
255.255.255.255               6   2,300 bytes          0      0 bytes               6       2,
300 bytes
0.0.0.0                       5   1,710 bytes          5      1,710 bytes           0       0
bytes
173.194.127.173               4   240 bytes            2      132 bytes             2       10
```

```
8 bytes
17.172.239.127                3   206 bytes            1      74 bytes              2       13
2 bytes
173.194.122.15                3   206 bytes            1      74 bytes              2       13
2 bytes
108.168.254.140               3   198 bytes            1      66 bytes              2       13
2 bytes
173.194.127.226               3   206 bytes            1      74 bytes              2       13
2 bytes
224.0.0.22                    3   180 bytes            0      0 bytes               3       18
0 bytes
224.0.0.2                     2   92 bytes             0      0 bytes               2       92
 bytes
17.172.232.98                 2   157 bytes            1      103 bytes             1       54
 bytes
54.251.179.109                2   132 bytes            0      0 bytes               2       13
2 bytes
173.194.122.23                2   132 bytes            0      0 bytes               2       13
2 bytes
10.0.0.1                      1   60 bytes             1      60 bytes              0       0
bytes
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ tshark -r "$DEC" -q -z conv,tcp | tee reports/tcp_conversations.txt
```

```
TCP Conversations
Filter:<No Filter>
                                                                |       ←      | |       →      |
| |     Total     |    Relative    |   Duration   |
                                                                | Frames  Bytes | | Frames  Bytes
| | Frames  Bytes |      Start     |              |
192.168.0.15:51007          ↔ 198.90.20.111:80                    1257 1,885 kB      978 73 kB
    2235 1,958 kB    113.588792000        45.4035
192.168.0.15:50959          ↔ 199.27.79.193:80                    1209 1,816 kB      959 71 kB
    2168 1,887 kB     21.074236000        34.8633
192.168.0.15:51008          ↔ 198.90.20.111:80                    1159 1,739 kB      947 70 kB
    2106 1,810 kB    113.588792000        46.0179
192.168.0.15:51001          ↔ 199.27.79.193:80                    1145 1,725 kB      927 66 kB
    2072 1,791 kB     63.066042000        26.6088
192.168.0.15:51009          ↔ 198.90.20.111:80                     687 1,028 kB      572 44 kB
    1259 1,072 kB    113.605175000        28.7983
192.168.0.15:51014          ↔ 198.90.20.111:80                     664 991 kB        566 44 kB
    1230 1,036 kB    114.020474000        25.0538
192.168.0.15:51006          ↔ 198.90.20.111:80                     654 970 kB        520 40 kB
    1174 1,011 kB    113.578039000        25.4942
192.168.0.15:51000          ↔ 199.27.79.193:80                     375 561 kB        323 25 kB
     698 586 kB       58.731642000         4.4552
192.168.0.15:50999          ↔ 199.27.79.193:80                     206 303 kB        180 15 kB
     386 319 kB       56.792636000         1.9390
192.168.0.15:51036          ↔ 192.168.0.9:5000                      98 144 kB         70 5,705 byt
es     168 150 kB      243.325633000         0.1777
192.168.0.15:50958          ↔ 199.27.79.193:80                      73 93 kB          67 12 kB
     140 105 kB       20.145401000        14.9846
192.168.0.15:51035          ↔ 192.168.0.9:5000                      80 116 kB         56 4,756 byt
es     136 121 kB      243.281602000         0.1398
192.168.0.15:50983          ↔ 173.194.127.141:80                    71 99 kB          59 4,392 byt
es     130 104 kB       27.831547000        31.4188
192.168.0.15:50997          ↔ 199.27.79.193:80                      62 87 kB          37 3,826 byt
es      99 90 kB        56.126972000        11.0492
192.168.0.15:51034          ↔ 173.194.127.229:443                   23 13 kB          32 9,731 byt
es      55 22 kB       234.384001000         6.7568
192.168.0.15:51005          ↔ 198.90.20.111:80                      25 22 kB          28 4,757 byt
es      53 27 kB       113.019956000         2.8111
192.168.0.15:50960          ↔ 31.13.70.1:443                        26 15 kB          25 3,151 byt
es      51 19 kB        21.329724000        32.8418
192.168.0.15:51028          ↔ 173.194.127.227:443                   18 11 kB          29 7,603 byt
es      47 18 kB       172.217100000        37.9452
192.168.0.15:51003          ↔ 198.41.208.140:80                     21 25 kB          23 2,718 byt
es      44 28 kB       102.780857000        32.8878
192.168.0.15:51018          ↔ 173.194.127.55:80                     20 24 kB          16 1,494 byt
es      36 26 kB       115.102391000        30.4286
192.168.0.15:50973          ↔ 199.16.158.186:443                    15 5,156 bytes      19 2,640 b
ytes      34 7,796 bytes    25.665085000        31.5233
192.168.0.15:51002          ↔ 104.74.158.12:80                      18 20 kB          15 1,236 byt
es      33 21 kB        63.175099000        30.4746
192.168.0.15:50974          ↔ 199.16.158.186:443                    14 5,082 bytes      18 2,574 b
ytes      32 7,656 bytes    25.665085000        31.5223
192.168.0.15:51016          ↔ 173.194.127.55:80                     13 15 kB          19 1,749 byt
es      32 16 kB       115.101879000        30.4439
192.168.0.15:51020          ↔ 74.125.203.156:80                     16 17 kB          16 2,534 byt
es      32 20 kB       115.146935000        31.5115
192.168.0.15:50956          ↔ 198.41.209.137:80                     13 14 kB          15 1,556 byt
es      28 16 kB        13.669693000        31.2704
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ tshark -r "$DEC" -Y 'ip' -T fields \
-e frame.number \
-e wlan.sa \
-e wlan.da \
-e ip.src \
-e ip.dst \
-e _ws.col.Protocol \
| head -n 200 | tee reports/ip_mac_mapping_sample.tsv
7                       0.0.0.0 255.255.255.255 DHCP
9                       0.0.0.0 255.255.255.255 DHCP
11                      192.168.0.1     255.255.255.255 DHCP
20                      192.168.0.15    8.8.8.8 DNS
21                      192.168.0.15    8.8.8.8 DNS
22                      192.168.0.15    224.0.0.2       IGMPv2
23                      192.168.0.15    224.0.0.251     IGMPv2
24                      192.168.0.15    224.0.0.2       IGMPv2
25                      192.168.0.15    224.0.0.251     IGMPv2
26                      8.8.8.8 192.168.0.15    DNS
27                      8.8.8.8 192.168.0.15    DNS
```

```
30                       192.168.0.15    17.172.239.31   TCP
31                       192.168.0.15    17.172.239.127  TCP
33                       17.172.239.31   192.168.0.15    TCP
35                       192.168.0.15    17.172.239.31   TCP
36                       192.168.0.15    17.172.239.31   TLSv1
37                       17.172.239.127  192.168.0.15    TCP
38                       192.168.0.15    17.172.239.127  TCP
42                       17.172.239.31   192.168.0.15    TCP
43                       17.172.239.31   192.168.0.15    TLSv1
44                       17.172.239.31   192.168.0.15    TLSv1
45                       17.172.239.31   192.168.0.15    TLSv1
47                       0.0.0.0 255.255.255.255 DHCP
48                       0.0.0.0 255.255.255.255 DHCP
49                       192.168.0.15    17.172.239.31   TCP
50                       192.168.0.15    17.172.239.31   TCP
51                       192.168.0.1     192.168.0.15    DHCP
52                       192.168.0.15    17.172.239.31   TLSv1
55                       17.172.239.31   192.168.0.15    TCP
56                       192.168.0.15    17.172.239.31   TLSv1
57                       17.172.239.31   192.168.0.15    TCP
58                       17.172.239.31   192.168.0.15    TLSv1
```

```
59                      192.168.0.15    17.172.239.31   TCP
60                      192.168.0.15    17.172.239.31   TLSv1
61                      17.172.239.31   192.168.0.15    TLSv1
65                      192.168.0.15    17.172.239.31   TCP
66                      192.168.0.15    17.172.239.31   TLSv1
69                      17.172.239.31   192.168.0.15    TCP
70                      192.168.0.15    17.172.239.31   TLSv1
74                      17.172.239.31   192.168.0.15    TCP
75                      17.172.239.31   192.168.0.15    TLSv1
78                      192.168.0.15    17.172.239.31   TCP
81                      192.168.0.15    224.0.0.251     MDNS
83                      192.168.0.15    224.0.0.251     MDNS
85                      192.168.0.15    224.0.0.251     MDNS
87                      192.168.0.15    224.0.0.251     MDNS
91                      192.168.0.15    224.0.0.251     MDNS
93                      192.168.0.15    224.0.0.251     MDNS
97                      192.168.0.15    8.8.8.8 DNS
98                      8.8.8.8 192.168.0.15    DNS
99                      192.168.0.15    198.41.209.137  TCP
100                     198.41.209.137  192.168.0.15    TCP
101                     192.168.0.15    198.41.209.137  TCP
```

```
102                     192.168.0.15    198.41.209.137  HTTP
103                     198.41.209.137  192.168.0.15    TCP
104                     192.168.0.15    224.0.0.251     MDNS
106                     192.168.0.15    224.0.0.251     MDNS
107                     198.41.209.137  192.168.0.15    HTTP
108                     198.41.209.137  192.168.0.15    TCP
109                     198.41.209.137  192.168.0.15    HTTP
110                     198.41.209.137  192.168.0.15    TCP
111                     198.41.209.137  192.168.0.15    HTTP
112                     198.41.209.137  192.168.0.15    HTTP
113                     198.41.209.137  192.168.0.15    HTTP
114                     198.41.209.137  192.168.0.15    HTTP
115                     198.41.209.137  192.168.0.15    HTTP
116                     198.41.209.137  192.168.0.15    TCP
117                     192.168.0.15    198.41.209.137  TCP
118                     192.168.0.15    198.41.209.137  TCP
119                     192.168.0.15    198.41.209.137  TCP
120                     192.168.0.15    198.41.209.137  TCP
121                     192.168.0.15    198.41.209.137  TCP
122                     192.168.0.15    198.41.209.137  TCP
123                     192.168.0.15    198.41.209.137  TCP
```

```
124                    192.168.0.15    198.41.209.137  TCP
125                    192.168.0.15    198.41.209.137  TCP
126                    192.168.0.15    198.41.209.137  TCP
127                    192.168.0.15    8.8.8.8 DNS
128                    8.8.8.8 192.168.0.15    DNS
129                    192.168.0.15    23.21.196.154   TCP
130                    23.21.196.154   192.168.0.15    TCP
131                    192.168.0.15    23.21.196.154   TCP
132                    192.168.0.15    23.21.196.154   HTTP
133                    192.168.0.15    23.21.196.154   TCP
134                    17.172.232.98   192.168.0.15    TLSv1
135                    192.168.0.15    17.172.232.98   TCP
136                    192.168.0.15    8.8.8.8 DNS
137                    8.8.8.8 192.168.0.15    DNS
138                    192.168.0.15    199.27.79.193   TCP
139                    192.168.0.15    199.27.79.193   TCP
140                    192.168.0.15    199.27.79.193   TCP
141                    192.168.0.15    199.27.79.193   HTTP
142                    199.27.79.193   192.168.0.15    TCP
143                    199.27.79.193   192.168.0.15    TCP
144                    199.27.79.193   192.168.0.15    TCP
```

```
145                    199.27.79.193   192.168.0.15    TCP
146                    199.27.79.193   192.168.0.15    TCP
147                    199.27.79.193   192.168.0.15    TCP
148                    199.27.79.193   192.168.0.15    TCP
149                    199.27.79.193   192.168.0.15    TCP
150                    199.27.79.193   192.168.0.15    TCP
151                    199.27.79.193   192.168.0.15    TCP
152                    199.27.79.193   192.168.0.15    TCP
153                    192.168.0.15    199.27.79.193   TCP
154                    192.168.0.15    199.27.79.193   TCP
155                    192.168.0.15    199.27.79.193   TCP
156                    192.168.0.15    199.27.79.193   TCP
157                    192.168.0.15    199.27.79.193   TCP
158                    192.168.0.15    199.27.79.193   TCP
159                    192.168.0.15    199.27.79.193   TCP
160                    192.168.0.15    8.8.8.8 DNS
161                    8.8.8.8 192.168.0.15    DNS
162                    192.168.0.15    199.27.79.193   TCP
163                    199.27.79.193   192.168.0.15    TCP
164                    192.168.0.15    199.27.79.193   TCP
165                    192.168.0.15    8.8.8.8 DNS
```

166	192.168.0.15	199.27.79.193	TCP
167	192.168.0.15	199.27.79.193	HTTP
168	8.8.8.8	192.168.0.15	DNS
169	192.168.0.15	31.13.70.1	TCP
170	199.27.79.193	192.168.0.15	TCP
171	199.27.79.193	192.168.0.15	TCP
172	199.27.79.193	192.168.0.15	TCP
173	199.27.79.193	192.168.0.15	TCP
174	199.27.79.193	192.168.0.15	TCP
175	199.27.79.193	192.168.0.15	TCP
176	199.27.79.193	192.168.0.15	TCP
177	199.27.79.193	192.168.0.15	TCP
178	199.27.79.193	192.168.0.15	TCP
179	199.27.79.193	192.168.0.15	TCP
180	192.168.0.15	199.27.79.193	TCP
181	192.168.0.15	199.27.79.193	TCP
182	192.168.0.15	199.27.79.193	TCP
183	192.168.0.15	199.27.79.193	TCP
184	192.168.0.15	199.27.79.193	TCP
185	31.13.70.1	192.168.0.15	TCP
186	192.168.0.15	31.13.70.1	TCP

187	192.168.0.15	31.13.70.1	TLSv1
188	199.27.79.193	192.168.0.15	TCP
189	199.27.79.193	192.168.0.15	TCP
190	199.27.79.193	192.168.0.15	TCP
191	199.27.79.193	192.168.0.15	TCP
192	199.27.79.193	192.168.0.15	TCP
193	199.27.79.193	192.168.0.15	TCP
194	199.27.79.193	192.168.0.15	TCP
195	199.27.79.193	192.168.0.15	TCP
196	192.168.0.15	199.27.79.193	TCP
197	199.27.79.193	192.168.0.15	TCP
198	192.168.0.15	199.27.79.193	TCP
199	192.168.0.15	199.27.79.193	TCP
200	192.168.0.15	199.27.79.193	TCP
201	192.168.0.15	199.27.79.193	TCP
202	199.27.79.193	192.168.0.15	TCP
203	199.27.79.193	192.168.0.15	TCP
204	199.27.79.193	192.168.0.15	TCP
205	192.168.0.15	199.27.79.193	TCP
206	192.168.0.15	199.27.79.193	TCP
207	192.168.0.15	199.27.79.193	TCP

```
208                     192.168.0.15    199.27.79.193   TCP
209                     192.168.0.15    199.27.79.193   TCP
210                     192.168.0.15    199.27.79.193   TCP
211                     192.168.0.15    199.27.79.193   TCP
212                     31.13.70.1      192.168.0.15    TCP
213                     31.13.70.1      192.168.0.15    TLSv1.2
214                     192.168.0.15    31.13.70.1      TCP
215                     192.168.0.15    31.13.70.1      TLSv1.2
216                     192.168.0.15    31.13.70.1      TLSv1.2
217                     192.168.0.15    31.13.70.1      TLSv1.2
218                     199.27.79.193   192.168.0.15    TCP
219                     199.27.79.193   192.168.0.15    TCP
220                     192.168.0.15    199.27.79.193   TCP
221                     192.168.0.15    199.27.79.193   TCP
222                     199.27.79.193   192.168.0.15    TCP
223                     199.27.79.193   192.168.0.15    TCP
224                     192.168.0.15    199.27.79.193   TCP
225                     199.27.79.193   192.168.0.15    TCP
226                     192.168.0.15    199.27.79.193   TCP
227                     199.27.79.193   192.168.0.15    TCP
228                     199.27.79.193   192.168.0.15    TCP
```

```
229                     199.27.79.193   192.168.0.15    TCP
230                     199.27.79.193   192.168.0.15    TCP
231                     192.168.0.15    199.27.79.193   TCP
232                     192.168.0.15    199.27.79.193   TCP
233                     192.168.0.15    199.27.79.193   TCP
234                     192.168.0.15    199.27.79.193   TCP
235                     192.168.0.15    199.27.79.193   TCP
236                     31.13.70.1      192.168.0.15    TCP
237                     31.13.70.1      192.168.0.15    TCP
238                     31.13.70.1      192.168.0.15    TCP
239                     199.27.79.193   192.168.0.15    TCP
240                     199.27.79.193   192.168.0.15    TCP
241                     199.27.79.193   192.168.0.15    TCP
242                     192.168.0.15    199.27.79.193   TCP
243                     199.27.79.193   192.168.0.15    TCP
244                     199.27.79.193   192.168.0.15    TCP
245                     199.27.79.193   192.168.0.15    TCP
246                     199.27.79.193   192.168.0.15    TCP
247                     199.27.79.193   192.168.0.15    TCP
248                     199.27.79.193   192.168.0.15    TCP
249                     199.27.79.193   192.168.0.15    TCP
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ ls -lh "$DEC"
-rw-rw-r-- 1 kali kali 13M Aug 11 18:58 working/file_working-dec

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ mkdir -p exported/http_objects exported/foremost

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ tshark -r "$DEC" --export-objects http,exported/http_objects 2>&1 | tee reports/http_expo
rt_log.txt
    1   0.000000 EFMNetworks_55:97:d4 → Broadcast    ARP 42 Who has 192.168.0.15? Tell 192.16
8.0.1
    2   0.943674 Apple_68:96:7c → NetisTechnol_48:e5:b4 ARP 42 Who has 192.168.1.1? Tell 192.
168.1.41
    3   0.944186 Apple_68:96:7c → NetisTechnol_48:a8:b5 ARP 42 Who has 192.168.0.1? Tell 192.
168.0.15
    4   0.944186 Apple_68:96:7c → 08:10:77:92:7c:2f ARP 42 Who has 192.168.1.1? Tell 192.168.
1.12
    5   0.944186 Apple_68:96:7c → EFMNetworks_55:97:d4 ARP 42 Who has 192.168.0.1? Tell 192.1
68.0.15
    6   0.944186 Apple_68:96:7c → TendaTechnol_58:c4:ba ARP 42 Who has 192.168.10.1? Tell 192
.168.10.113
    7   0.944185      0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Request  - Transaction ID 0×21e
6be1c
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ find exported/http_objects -type f -ls
   526566      4 -rw-r--r--   1 kali     kali         2671 Aug 11 19:26 exported/http_objects
/logo.png
   526535      4 -rw-r--r--   1 kali     kali          129 Aug 11 19:26 exported/http_objects
/&i=19612&llnocookies=undefined
   526473    180 -rw-r--r--   1 kali     kali       181022 Aug 11 19:26 exported/http_objects
/hit.json
   526556      4 -rw-r--r--   1 kali     kali         3928 Aug 11 19:26 exported/http_objects
/rss.png
   526538     16 -rw-r--r--   1 kali     kali        12768 Aug 11 19:26 exported/http_objects
/qpy-UiLNKP-VfOdbcs6r6-vvDin1pK8aKteLpeZ5c0A.woff
   526493      4 -rw-r--r--   1 kali     kali          166 Aug 11 19:26 exported/http_objects
/adc-i-00aecd.png
   526523      8 -rw-r--r--   1 kali     kali         4862 Aug 11 19:26 exported/http_objects
/superfish.min.js
   526553      4 -rw-r--r--   1 kali     kali         1226 Aug 11 19:26 exported/http_objects
/category-strip.png
   526530     12 -rw-r--r--   1 kali     kali        12076 Aug 11 19:26 exported/http_objects
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ foremost -i "$DEC" -o exported/foremost | tee reports/foremost_log.txt
Processing: working/file_working-dec
|*|

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ find exported/foremost -type f -ls
   526613      4 -rw-rw-r--   1 kali     kali         1604 Aug 11 19:27 exported/foremost/aud
it.txt
   526614    260 -rw-rw-r--   1 kali     kali       264865 Aug 11 19:27 exported/foremost/jpg
/00000104.jpg
   526615     88 -rw-rw-r--   1 kali     kali        89458 Aug 11 19:27 exported/foremost/jpg
/00004538.jpg
   526619      4 -rw-rw-r--   1 kali     kali           35 Aug 11 19:27 exported/foremost/gif
/00003653.gif
   526618      4 -rw-rw-r--   1 kali     kali           42 Aug 11 19:27 exported/foremost/gif
/00002450.gif
   526617      4 -rw-rw-r--   1 kali     kali           35 Aug 11 19:27 exported/foremost/gif
/00001260.gif
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ find exported/foremost -maxdepth 2 -type d -print
exported/foremost
exported/foremost/jpg
exported/foremost/gif
exported/foremost/png
exported/foremost/htm

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ find exported -type f -exec file {} \; | tee reports/exported_file_types.txt
exported/foremost/audit.txt: ASCII text
exported/foremost/jpg/00000104.jpg: JPEG image data, JFIF standard 1.01, aspect ratio, densit
y 1×1, segment length 16, baseline, precision 8, 2492×2492, components 3
exported/foremost/jpg/00004538.jpg: JPEG image data, JFIF standard 1.01, resolution (DPI), de
nsity 72×72, segment length 16, comment: "Lavc55.69.100", baseline, precision 8, 718×404, com
ponents 3
exported/foremost/gif/00003653.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00002450.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00001260.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00000406.gif: GIF image data, version 89a, 1 x 1
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ find exported -type f -exec sha256sum {} \; | tee reports/exported_file_hashes.txt
d54f38f5df345950c96276c918d553695e4bd504c3c4c808324b971b576f7265  exported/foremost/audit.txt
3f9bed324e04c1735b498dab0f138673e20591a5b39201e731a91dd180929ba6  exported/foremost/jpg/00000
104.jpg
f891fa7ef7652a26a54638dd7b0aebcb60676fd72ec42a2a636523a6371297dc  exported/foremost/jpg/00004
538.jpg
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015  exported/foremost/gif/00003
653.gif
ef1955ae757c8b966c83248350331bd3a30f658ced11f387f8ebf05ab3368629  exported/foremost/gif/00002
450.gif
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015  exported/foremost/gif/00001
260.gif
a0d3a0aff7dc3bf32d2176fc3dcda6e7aba2867c4f4d1f7af6355d2cfc6c44f8  exported/foremost/gif/00000
406.gif
53f7e56783c414746a9fc36126375095604246l7e3135248640eac142d6e0f94  exported/foremost/gif/00010
456.gif
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015  exported/foremost/gif/00003
719.gif
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015  exported/foremost/gif/00010
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ find exported -type f -ls
   526613      4 -rw-rw-r--   1 kali     kali         1604 Aug 11 19:27 exported/foremost/aud
it.txt
   526614    260 -rw-rw-r--   1 kali     kali       264865 Aug 11 19:27 exported/foremost/jpg
/00000104.jpg
   526615     88 -rw-rw-r--   1 kali     kali        89458 Aug 11 19:27 exported/foremost/jpg
/00004538.jpg
   526619      4 -rw-rw-r--   1 kali     kali           35 Aug 11 19:27 exported/foremost/gif
/00003653.gif
   526618      4 -rw-rw-r--   1 kali     kali           42 Aug 11 19:27 exported/foremost/gif
/00002450.gif
   526617      4 -rw-rw-r--   1 kali     kali           35 Aug 11 19:27 exported/foremost/gif
/00001260.gif
   526616      4 -rw-rw-r--   1 kali     kali           35 Aug 11 19:27 exported/foremost/gif
/00000406.gif
   526621      4 -rw-rw-r--   1 kali     kali           49 Aug 11 19:27 exported/foremost/gif
/00010456.gif
   526620      4 -rw-rw-r--   1 kali     kali           35 Aug 11 19:27 exported/foremost/gif
/00003719.gif
```

```
┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ ls -lh reports/http_export_log.txt \
      reports/foremost_log.txt \
      reports/exported_file_types.txt \
      reports/exported_file_hashes.txt
-rw-rw-r-- 1 kali kali  17K Aug 11 19:29 reports/exported_file_hashes.txt
-rw-rw-r-- 1 kali kali  14K Aug 11 19:29 reports/exported_file_types.txt
-rw-rw-r-- 1 kali kali    0 Aug 11 19:27 reports/foremost_log.txt
-rw-rw-r-- 1 kali kali 2.3M Aug 11 19:26 reports/http_export_log.txt

┌──(kali㉿kali)-[~/CIP-B103-Lab9]
└─$ find exported -type f -ls
cat reports/exported_file_types.txt
cat reports/exported_file_hashes.txt
   526613      4 -rw-rw-r--   1 kali     kali         1604 Aug 11 19:27 exported/foremost/aud
it.txt
   526614    260 -rw-rw-r--   1 kali     kali       264865 Aug 11 19:27 exported/foremost/jpg
/00000104.jpg
   526615     88 -rw-rw-r--   1 kali     kali        89458 Aug 11 19:27 exported/foremost/jpg
/00004538.jpg
```